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Business 510

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Predicting CO_2 Emissions by Vehicles using Regression

Problem Statement

Transportation is one of the largest contributors to greenhouse gas emissions. Environmental policy is a complex challenge faced by numerous stakeholders globally, including governments, automative manufacturers, and environmental agencies. Consumers are growing increasingly conscious of the impact of their individual choices on the environment. The objective of this project is to use multiple linear regression to create a model to predict carbon dioxide emissions based on a series of vehicle-specific characteristics and understand which causes the largest impact.

Data Collection Source

Data was accessed through Kaggle but was originally published by the Canadian Government as part of their natural resources and environmental reporting. It contains variables such as make, car model, vehicle class, fuel types, engine characteristics, fuel consumption ratings, and certified CO_2 emissions for multiple vehicle models. The data was clean but required dummy variable creation, elimination of redundant predictors, and randomized splitting for cross-validation.

Variable Selection

Dependent Variable (Y)

CO_2 Emissions (g/km) — actual emissions measured by regulatory agencies.

Independent Variables (X)

Fuel Consumption Combined (L/100 km). This variable indicates the fuel consumption combined on highways and in cities. (Individual values for highway & city mentioned in the data sheet.) Strongest theoretical predictor of emissions (direct relationship).

Fuel Consumption Comb (L/100 km)	CO2 Emissions(g/km)
10.1	232
11	259
13.3	306

Engine Size (L). Assumption is larger engines burn more fuel which leads to more emissions.

Transmission Type (Dummy Variables). The following values are categorized to Automatic, Manual, or Semi-Automatic. They were then turned into dummy variables

- A6, A8 → Automatic becomes baseline.
- M5, M6, M7 → Trans_Manual (1 = manual, 0 = otherwise).
- AM6, AS5, AV7 → Trans_SemiAuto (1 = SemiAuto, 0 = otherwise).

Fuel Type (Dummy Variable). The following Fuel Type variables were converted into dummy variables: Fuel Type_D, Fuel Type_E, Fuel Type_Z, Fuel Type_X. Fuel Type_N was the baseline for the model.

- D → Diesel
- E → Ethanol (E85)
- N → Natural Gas
- X → Premium Gasoline
- Z → Regular Gasoline

Engine Size(L)	Cylinders	Transmission	Fuel Type
2	4	AS5	Z
2.4	4	M6	Z
1.5	4	AV7	Z

Cylinders	Engine Size(L)	Trans_manual	Trans_semiauto	Fuel Type_Z	Fuel Type_D	Fuel Type_X	Fuel Type_E
5	2.5	0	1	0	0	1	0
6	3	0	1	1	0	0	0
8	5	0	1	1	0	0	0

Tests: Significance & Normality

Significance F < 0.001 is less than alpha of 0.05. Therefore, the model is Significant as a whole. P-values are less than alpha of 0.05 for all included independent variables except Transmission Semi-Automatic. All other variables are Significant.

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	17201137.007	7	2457305.287	82471.808	<.001[b]
	Residual	153775.610	5161	29.796		
	Total	17354912.617	5168			

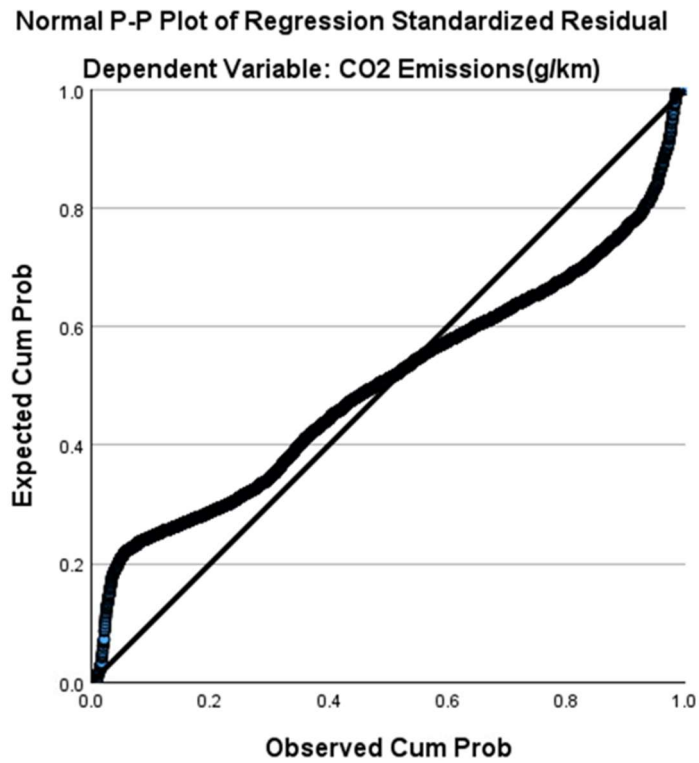
a. Dependent Variable: CO2 Emissions(g/km)

b. Predictors: (Constant), Fuel Consumption Comb (L/100 km), Fuel Type_D, Trans_semiauto, Fuel Type_Z, Fuel Type_E, Trans_manual, Engine Size(L)

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
		B	Std. Error	Beta			Tolerance	VIF
1	(Constant)	7.083	.443		16.002	<.001		
	Engine Size(L)	.861	.110	.020	7.840	<.001	.262	3.817
	Trans_manual	-.734	.263	-.005	-2.785	.005	.608	1.644
	Trans_semiauto	-.200	.199	-.002	-1.007	.314	.603	1.659
	Fuel Type_Z	.484	.168	.004	2.878	.004	.831	1.203
	Fuel Type_D	30.348	.525	.077	57.851	<.001	.963	1.039
	Fuel Type_E	-111.624	.448	-.424	-249.146	<.001	.593	1.688
	Fuel Consumption Comb (L/100 km)	22.391	.058	1.103	385.489	<.001	.210	4.769

a. Dependent Variable: CO2 Emissions(g/km)



Tests: Multicollinearity

No independent variables in our data exhibit high or moderate multicollinearity (all VIF < 5.000). An earlier proposal included number of engine cylinders, but this caused multicollinearity with the Engine Size variable, so Cylinders was discarded.

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.	Collinearity Statistics Tolerance	Collinearity Statistics VIF
1	(Constant)	7.083	.443		16.002	<.001		
	Engine Size(L)	.861	.110	.020	7.840	<.001	.262	3.817
	Trans_manual	-.734	.263	-.005	-2.785	.005	.608	1.644
	Trans_semiauto	-.200	.199	-.002	-1.007	.314	.603	1.659
	Fuel Type_Z	.484	.168	.004	2.878	.004	.831	1.203
	Fuel Type_D	30.348	.525	.077	57.851	<.001	.963	1.039
	Fuel Type_E	-111.624	.448	-.424	-249.146	<.001	.593	1.688
	Fuel Consumption Comb (L/100 km)	22.391	.058	1.103	385.489	<.001	.210	4.769

a. Dependent Variable: CO2 Emissions(g/km)

Predictive Analytics Model and Analysis

Hypothesis Testing

H_0: Independent variables do NOT significantly predict CO_2 emissions.

H_A: At least one variable significantly predicts CO_2 emissions.

Significant Predictors ($p < 0.05$): Fuel Consumption (strongest), Engine Size, Fuel Type D, Fuel Type E, Transmission Manual ($p = .005$)

Insignificant Predictors: Transmission SemiAuto

Result: Reject H_0. Variables do significantly predict emissions.

Results with Cylinder & Fuel type X:

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.	Collinearity Statistics Tolerance	Collinearity Statistics VIF
1	(Constant)	-74.810	5.357		-13.964	<.001		
	Cylinders	.533	.117	.017	4.554	<.001	.121	8.237
	Engine Size(L)	.269	.167	.006	1.610	.107	.108	9.227
	Trans_manual	-.746	.257	-.005	-2.897	.004	.608	1.645
	Trans_semiauto	-.259	.196	-.002	-1.321	.186	.595	1.681
	Fuel Type_Z	81.578	5.332	.697	15.299	<.001	.001	1269.106
	Fuel Type_D	111.461	5.357	.284	20.808	<.001	.009	113.564
	Fuel Type_X	81.317	5.333	.702	15.249	<.001	.001	1293.339
	Fuel Type_E	-30.157	5.344	-.115	-5.643	<.001	.004	251.868
	Fuel Consumption Comb (L/100 km)	22.355	.058	1.101	386.974	<.001	.202	4.948

a. Dependent Variable: CO2 Emissions(g/km)

Results eliminating Cylinder & Fuel Type X:

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.	Collinearity Statistics Tolerance	Collinearity Statistics VIF
1	(Constant)	7.083	.443		16.002	<.001		
	Engine Size(L)	.861	.110	.020	7.840	<.001	.262	3.817
	Trans_manual	-.734	.263	-.005	-2.785	.005	.608	1.644
	Trans_semiauto	-.200	.199	-.002	-1.007	.314	.603	1.659
	Fuel Type_Z	.484	.168	.004	2.878	.004	.831	1.203
	Fuel Type_D	30.348	.525	.077	57.851	<.001	.963	1.039
	Fuel Type_E	-111.624	.448	-.424	-249.146	<.001	.593	1.688
	Fuel Consumption Comb (L/100 km)	22.391	.058	1.103	385.489	<.001	.210	4.769

a. Dependent Variable: CO2 Emissions(g/km)

Training Model Results

R^2 was 0.991, meaning 99.1% of the variation in dependent variable, CO_2 Emissions, is explained by the seven independent variables.

Estimated Multiple Linear Regression Equation: Y = 7.083 + 0.861(xEngine Size) - 0.734 (xTrans_Manual) - 0.200(xTrans_Semiauto) + 0.484(xFuel Type_Z) + 30.348(xFuel Type_D) - 111.624(xFuel Type_E) + 22.391(xFuel Consumption Comb)

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.	Collinearity Statistics Tolerance	Collinearity Statistics VIF
1	(Constant)	7.083	.443		16.002	<.001		
	Engine Size(L)	.861	.110	.020	7.840	<.001	.262	3.817
	Trans_manual	-.734	.263	-.005	-2.785	.005	.608	1.644
	Trans_semiauto	-.200	.199	-.002	-1.007	.314	.603	1.659
	Fuel Type_Z	.484	.168	.004	2.878	.004	.831	1.203
	Fuel Type_D	30.348	.525	.077	57.851	<.001	.963	1.039
	Fuel Type_E	-111.624	.448	-.424	-249.146	<.001	.593	1.688
	Fuel Consumption Comb (L/100 km)	22.391	.058	1.103	385.489	<.001	.210	4.769

a. Dependent Variable: CO2 Emissions(g/km)

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.996[a]	.991	.991	5.459

a. Predictors: (Constant), Fuel Consumption Comb (L/100 km), Fuel Type_D, Trans_semiauto, Fuel Type_Z, Fuel Type_E, Trans_manual, Engine Size(L)

b. Dependent Variable: CO2 Emissions(g/km)

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.9956
R Square	0.9911
Adjusted R Square	0.9911
Standard Error	5.4585
Observations	5169.0000

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	7	17201137.007	2457305.287	82471.808	0
Residual	5161	153775.610	29.796		
Total	5168	17354912.617			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	7.083	0.443	16.002	0.000	6.216	7.951	6.216	7.951
Engine Size(L)	0.861	0.110	7.840	0.000	0.646	1.076	0.646	1.076
Trans_manual	-0.734	0.263	-2.785	0.005	-1.250	-0.217	-1.250	-0.217
Trans_semiauto	-0.200	0.199	-1.007	0.314	-0.591	0.190	-0.591	0.190
Fuel Type_Z	0.484	0.168	2.878	0.004	0.154	0.813	0.154	0.813
Fuel Type_D	30.348	0.525	57.851	0.000	29.320	31.377	29.320	31.377
Fuel Type_E	-111.624	0.448	-249.146	0.000	-112.502	-110.745	-112.502	-110.745
Fuel Consumption Comb (L/100 km)	22.391	0.058	385.489	0.000	22.277	22.505	22.277	22.505

Validation Model Results

R^2 had little variance between the training and validation model.

- Training R^2: 99.10%
- Validation R^2: 99.08%

RMSE had little variance between the training and validation model.

- Training RMSE: 5.4543
- Validation RMSE: 5.7226

Standard Error had little variance between the training and validation model.

- Training SE: 5.459
- Validation SE: 5.733

CO2 Emissions(g/km)	Predicted CO2 Emissions(g/km)	Residuals	Residuals^2	$y - \bar{y}$	$(y - \bar{y})^2$
254	254.442	-0.442	0.195	2.943	8.662
302	301.922	0.078	0.006	50.943	2595.204
275	271.382	3.618	13.087	23.943	573.274

SST
7925230.84

SSE
72570.52

R^2
0.9908

SE
5.7330

MSE
32.748

RMSE Validation
5.7226

SST
7925230.84

SSE
72570.52

R^2
0.9908

SE
5.7330

MSE
32.748

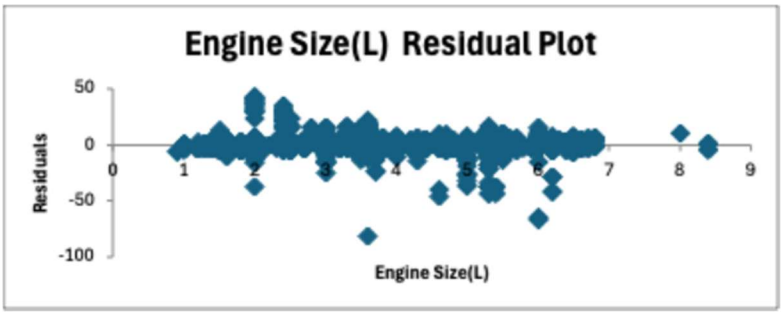
RMSE Validation
5.7226

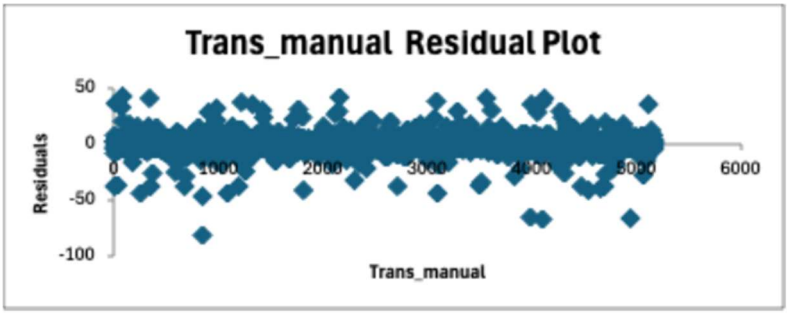
RMSE Training
5.4543

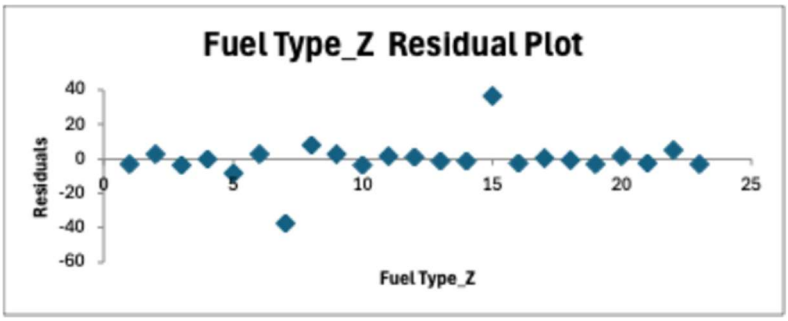
Conclusion: The validation model suggests the estimated multiple linear regression equation is a good fit to explain the variation in CO_2 emissions.

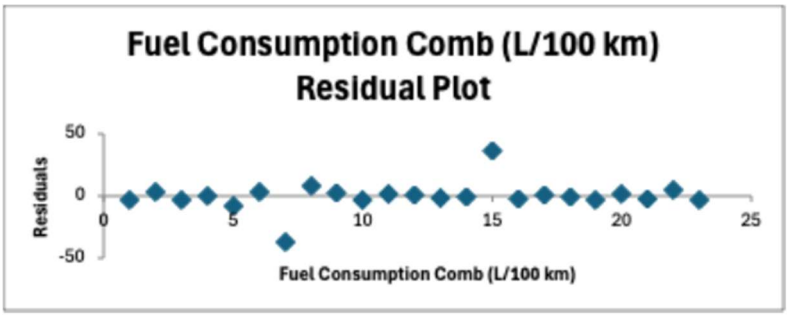
Analysis of Residuals

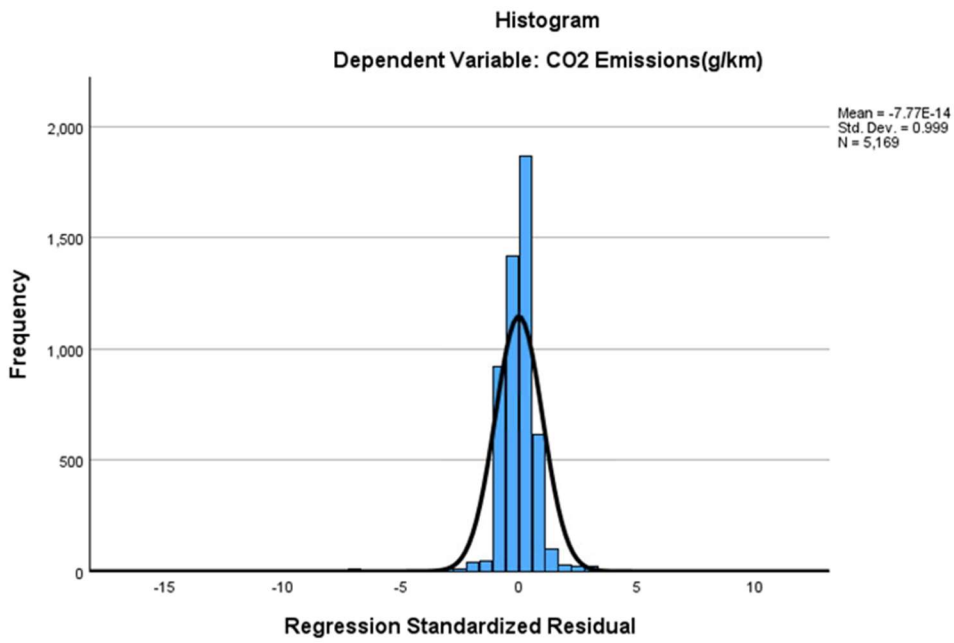
- Linearity Holds: random dispersion of residuals around horizontal axis.
- Homoscedasticity Holds: constant spread around horizontal axis.
- Normality Holds: histogram of residuals follows bell-shaped curve.
- Independence Holds: no cyclical or alternating pattern of residuals.


Engine Size(L) Residual Plot
Residuals
50
0
-50
-100
0
1
2
3
4
5
6
7
8
9
Engine Size(L)


Trans_manual Residual Plot
Residuals
50
0
-50
-100
0
1000
2000
3000
4000
5000
6000
Trans_manual


Fuel Type_Z Residual Plot
Residuals
40
20
0
-20
-40
-60
0
5
10
15
20
25
Fuel Type_Z


Fuel Consumption Comb (L/100 km) Residual Plot
Residuals
50
0
-50
0
5
10
15
20
25
Fuel Consumption Comb (L/100 km)



Conclusion and Lessons Learned

Fuel consumption is the strongest predictor of CO_2 emissions. Vehicles that burn more fuel emit substantially more CO_2. Engine Size contributes, but less than expected. Larger engines increase emissions slightly. Fuel Type changes emissions dramatically. Diesel (D): Higher emissions than baseline fuel (X). Ethanol/E-fuel (E): Significantly lower emissions. Fuel chemistry impacts CO_2 more than the transmission. Transmission type has minimal environmental impact- Manual transmissions reduce emissions slightly. Emissions relationships are naturally linear- CO_2 output aligns closely with physical combustion principles. Fuel Consumption has a direct linear relationship with CO_2, that's why R^2 is naturally very high. Data preprocessing often matters more than modelling. Fixing multicollinearity, removing redundant predictors, and creating proper categories are the steps which improved the model.

Future Directions

If there was more time to explore, test, and analyze the dataset, we would explore the impact of removing and swapping independent variables from the model to analyse the impact

on R2 because of the very high ratio. Some opportunities for further analysis include swapping dummy variables, swapping Engine Size variable for Cylinder variable, and removing xTrans_SemiAuto from model due to being identified as an insignificant variable.